

Project Phase 1: Streamly Project Planning Submission

I. Introduction Details

Application Name: Streamly

Primary Customer: Daniyal

Next Meeting Date: 6 March 2026

Team Members:

- Devin Boodoo
- Jonathan Lavoie
- Maryam Hameed
- Ashwin Sivasakthi
- Avis Shrestha

II. Project Overview

2.1: Project Purpose

Streamly is a music library catalogue system inspired by platforms such as Spotify and Youtube Music. The purpose of the system is to provide users with an organized, filterable and searchable music database while allowing administrators full control over managing the underlying music and song records.

Streamly's Phase 1 heavily focuses on:

- Having structured authentication (admin and user)
- Powerful search and filtering capabilities
- A scalable music database
- Personalized engagement via favourites

Overall, the projected application emphasizes database-driven design, usability, and scalability.

2.2: Project Scope

The scope of this application project includes the following:

1. Core System
 - Music metadata database (components such as title, artist, genre, release year, popularity).
 - Role-based authentication (differentiating admin and user)
2. User Capabilities
 - Register and login
 - Filter by genre, year, popularity
 - Search by title, artist, album

- Favouritizing songs into a playlist
 - Search within favourites
3. Admin Capabilities
 - Adding songs
 - Editing metadata
 - Removing songs
 4. Out of Scope (Phase 1)
 - Users uploading their own music
 - Copyright licensing
 - Advanced audio analytics

2.3: High-Level Goals

1. Delivering a stable and secure database backbone
2. Provide fast and flexible search functionality
3. Maintain clean role separation between user and admin
4. Following agile planning principles
5. Delivering iterative progress across the three development cycles

III. Team Agreements & Elicitation Documentation

3.1 Team Contract Summary

1. Roles and Responsibilities

Member Name	Official Title	Title Description	Sub-Responsibilities
Devin	Technical Manager	Bridge management and development by making high-level technical decisions, and ensuring the team follows normal engineering practices	S1 Registration S2 Login S3 Logout S17 Play Music
Johnathan	Back-End Lead	Lead the design and development of the server, database, and API architecture that will power the application	S5 Searching by Title S6 Search Artist S7 Search Album S19 Searching Playlist
Maryam	Software Quality Lead	Plan the testing strategy and quality assurance process to ensure the application meets the functional and performance standards	S8 Genre Filtering S12 Favouritizing S13 Viewing Favourites S14 Adding Songs

Ashwin	Front-End Lead	Direct the design and implementation of the user interface and client-side experience	S4 Admin Login S15 Editing Songs S16 Removing Songs S18 Password Encryption
Avis	Project Manager	Oversee project planning, scope, timelines, and team coordination to ensure successful delivery	S9 Filter by year S10 Filter by trend S11 Trend Filtering

***see section 4 for details

2. Expectations

- Weekly 1 hour 30 minute meetings
- Clear GitHub commits
- Peer reviewing before merging
- Equal participation

3. Conflict Resoltuion

- Internal discussions
- Majority vote
- Instructor intervention if needed

3.2 Elicitation Methods Used

- Customer interview
- Structured Q&A
- Scope confirmation session
- Industry research
- Stakeholder identification
- Glossary creation
- Knowledge gap identification

3.3 Key Findings from Customer Meetings

1. Confirmed Customer Vision:
 - Two interfaces: Admin & User
 - Search bar must be flexible
 - Filtering by genre, popularity and date
 - Admin must manage database
 - Users cannot add songs
 - Favourite playlist functionality

2. Constraints

- Timeline: full semester
- System must be scalable
- Must use real music metadata

3. Identified Pain Points

- Users want more flexible search recognition
- Existing platforms have strict matching

IV. User Stories & Prioritization

4.1 Structured User Stories

1. Authentication Domain

Feature	User Story
S1 Registration	As a new user, I want to create an account, so that I can login and access personalized features.
S2 Login	As a registered user, I want to log into my account securely, so that I can access my library and favourites.
S3 Logout	As a logged-in user, I want to be able to log out, so that my account remains secure.
S4 Admin Login	As an admin, I want to log in with my admin credentials so that I can personally manage the system database.

2. Filtering & Search Domain (high priority)

Feature	User Story
S5 Search by song title	As a user I want to be able to search for a song by title, so that I can instantly find music I know.
S6 Search by artist name	As a user, I want to be able to search for a song by the artists' name, so that I can explore all songs by that artist.
S7 Search by album	As a user I want to be able to search songs by album so that I can find the song I'm looking for as well as the album its part of.
S8 Filter by genre	As a user, I want to filter songs by the genre, so that I can browse songs based on the mood.
S9 Filter by year	As a user, I want to filter songs by the release date, so that I can explore music from specific time periods.

S10 Filter by trend	As a user I want to filter songs by popularity so that I can know the trending songs.
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3. Playlist & Favourites

Feature	User Story
S11 Trend Filtering	As a user, I want to favourite a song, so that it gets added to my “jammed” playlist.
S12 Favouritizing	As a user, I want to view all of my favourite songs, so that I can quickly access my preferred music.
S13 Viewing Favourites	As a user, I want to be able to search inside my favourites, so that I can quickly locate a saved track.

4. Admin Functionality

Feature	User Story
S14 Adding Songs	As an admin, I want to add new songs to the database, so that all users can have access to real music.
S15 Editing Songs	As an admin, I want to edit song information (year, genre, artist) so that the data base information stays accurate.
S16 Removing Songs	As an admin, I want to be able to remove songs, or song information from the system, so that outdated, controversial or incorrect content can be deleted.

5. Nice to Haves

Feature	User Story
S17 Play Music	As a user, I want to play real music inside the app, so that I can stream songs directly.
S18 Password Encryption	As a user, I want my password encrypted so that my account remains protected.
S19 Searching Playlist	As a user, I want to be able to search within my playlist to find the song fasting

4.2 Prioritization

Priority Level	Feature Tags	Justification
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Must Haves	S1-S5 S14-16	System backbone and database control.
Should Haves	S6-S13	Improves usability and engagement
Nice to Haves	S17-S19	Enhancements beyond MVP

V. Effort Estimation & Iteration Planning

5.1 Planning Poker Estimates (in hours)

User Story	Estimated Working Hours	Justification	Task Assignment
S1 Registration	6 hours	Form validation and database integration	Devin
S2 Login Process	4 hours	Authentication and session handling	Devin
S3 Logout Process	4 hours	Session Handling	Devin
S4 Admin Login	2 hours	Role-based access	Ashwin
S5 Search by song title	4 hours	Query optimization	Jonathan
S6 Search by artist name	4 hours	Database linking	Jonathan
S7 Search by album	4 hours	Database linking	Jonathan
S8 Filter by genre	8 hours	Keyword/partial matching	Maryam
S9 Filter by year	2 hours	Query filter	Avis
S10 Filter by trend	2 hours	Query filter	Avis
S11 Trend Filtering	2 hours	Using popularity metrics	Avis
S12	2 hours	Mapping	Maryam

Favouritizing			
S13 Viewing Favourites	4 hours	Display logic	Maryam
S14 Adding Songs	2 hours	Filtering within a subset (playlist)	Maryam
S15 Editing Songs	4 hours	Admin User Interface and Database	Ashwin
S16 Removing Songs	4 hours	Updating Logic	Ashwin
S17 Play Music	8 hours	Audio player UI(play, pause, skip), Loading time errors,	Devin
S18 Password Encryption	2 hours	Hashing and adding a security layer	Ashwin
S19 Searching Playlist	2 hours	Filtering within a subset	Jonathan

5.2 Iteration Breakdown

Iteration 1 – Core Infrastructure

User Stories:

- S1-S4
- S14-S16

Goal: Functional authentication + working database

Deliverable: Fully operational admin - controlled database

Iteration 2 – Search & Filtering

User Stories:

- S5-S10

Goal: Complete browsing and flexible search functionality

Deliverable: Users can search and filter efficiently

Iteration 3 – Engagement & Security

User Stories:

- S11-S13
- S18

- S17

Goal: Personalization & system refinement
Deliverable: Favourites & encryption implemented.

5.3 Agile Board Representation

Title	Status	Assignees	Linked pull requests	Sub-issues progress
No Priority 18 Estimate: 0				
1 S1 registration #1	Backlog	Focusucuf		
2 S2: Login #2	Backlog	Focusucuf		
3 S3 Logout #3	Backlog	Focusucuf		
4 S11 Trend Filtering (2 Hours) #4	Backlog	AvisS12		
5 S4 Admin Login #5	Backlog	Ashwins1971		
6 S12 Favouritizing (2 Hours) #6	Backlog	maryammarji		
7 S5 Searching by Title #7	Backlog	Johnathan-Lavoie		
8 S13 Viewing Favourites (4 Hours) #8	Backlog	maryammarji		
9 S6: Search Artist #9	Backlog	Johnathan-Lavoie		
10 S7: Search Album #11	Backlog	Johnathan-Lavoie		
11 S14 Searching Playlist (2 Hours) #13	Backlog	Johnathan-Lavoie		
12 S15 Adding Songs #10	Backlog	maryammarji		
13 S16 Editing Songs #12	Backlog	Ashwins1971		
14 S17 Play Music #14	Backlog	Focusucuf		
15 S8: Flexible Search #16	Backlog	AvisS12		
16 S18 Password Encryption #15	Backlog	Ashwins1971		
17 S9: Genre Filtering #17	Backlog	maryammarji		
18 S10: Year Filtering #18	Backlog	AvisS12		

VI. Finalized Project Roadmap

6.1 Major Milestones

Weeks 1-2	- Database schema complete - Authentication system complete
Weeks 3-4	- Admin interface finalized (create, read,upload,delete) - Search functionality operational
Weeks 5-6	- Filtering complete - Playlist system integrated

	• Security enhancements • Testing and debugging
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6.2 Key Risks & Mitigation

Database scaling	Proper normalization
Query performance	Indexed fields
Integration conflicts	Weekly merging testing
Underestimating times	Buffering (esp in iteration 3)

6.2 Next Steps

- Finalize diagrams
- Initalized Github project board
- Begin iteration 1 development
- Implement authentication